

Deep-Learning-Based AoA and AoD Estimation in Analog Millimeter Wave MIMO Systems

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Abstract—Sixth-generation wireless technologies (6G) are expected to continue using millimeter wave (mmWave) frequencies, extending the spectrum usage to the Terahertz frequency range. High-accuracy angle of arrival (AoA) and angle of departure (AoD) estimation in mmWave channels is critical for cell search, stable communications, and positioning. Moreover, the design of low-complexity AoA/AoD estimation algorithms is also of major importance in the deployment of practical systems to enable a fast and resource-efficient computation of beamforming weights. Parametric mmWave channel estimation allows to describe the channel matrix as a combination of direction-dependent signal paths, exploiting the sparse characteristics of mmWave channels. Previous works have shown that the problem of estimating the AoA/AoD of such signal paths is equivalent to estimating the frequencies of sinusoidal components observed after a pilot-based training stage. In this work, we propose two adapted convolutional neural networks to address the AoA/AoD estimation: a residual neural network (ResNet) and a U-Net neural network. Assuming that the estimation problem can be seen as a supervised image-to-image translation problem, we have redesigned these basic architectures including a posterior peak search stage based on blob detection principles. Numerical simulations were performed to compare these neural networks with state-of-the-art signal processing techniques and the Cramér-Rao lower bound, including the analysis of their robustness in front of hardware impairments. Our results show that the proposed deep-learning-based schemes outperform significantly other approaches, especially in terms of probability of detection, while showing a competitive complexity in massive MIMO settings.

Index Terms—mmWave, channel estimation, analog beamforming, neural networks, ResNet, U-Net.

I. INTRODUCTION

THE use of millimeter wave (mmWave) frequencies is one of the key enablers of advanced functionalities in Beyond 5G (B5G) and 6G networks [1], [2], [3]. Although research is already focused on modeling and characterizing Terahertz frequency communications, mmWave bands will continue to

play a fundamental role in the deployment of enhanced mobile broadband services (eMBB). Communications in mmWave bands need to tackle phenomena such as large propagation losses and atmospheric absorption. Due to this, beamforming techniques that involve highly directional beams are deployed, usually accompanied by massive antenna arrays. A fundamental mechanism to allow such advanced adaptive arrays is the beam sweeping, measurement and reporting of the direction of transmission (characterized by the angle of arrival (AoA) and angle of departure (AoD) of the transmitter (Tx) and receiver (Rx) beams) of each potential link [4], [5].

6G technologies open the door to advanced communication features and applications, which rely on channel estimation [6]. For example, estimation of the AoA and AoD of the channel paths is fundamental for any localization and sensing method, including initial alignment [7] or wireless localization and tracking [8]. Parametric mmWave multiple input multiple output (MIMO) channel models have been used as a basis to derive efficient AoA and AoD estimation methods based on predefined codebooks of directions (identified by a pair of beamforming and combining vectors) that cover the entire angular space between Tx and Rx nodes. An efficient search among the candidate beam directions can be performed through hierarchical codebooks, combining an initial coarse beam search with subsequent finer ones [9], [10].

The mmWave MIMO channel model is sparse by nature [11], which is a central feature in the design of channel estimation solutions. Schemes relying on compressed sensing (CS) [12], such as the methods based on orthogonal matching pursuit (OMP) [13], [14], or least squares (LS)-based approaches [12], [15] have also received significant attention. An iterative algorithm for analog mmWave channel estimation in the transformed spatial domain, referred to as transformed spatial domain channel estimation (TSDCE), which outperforms both the LS and OMP schemes, has been proposed in [16], [17]. The TSDCE algorithm uses a discrete Fourier transform (DFT)-based codebook of analog beamforming (ABF) vectors, which are ordered so that the observation matrix matches a two-dimensional DFT (2D-DFT) of a sum of complex sinusoids in additive white Gaussian noise (AWGN). In the transformed spatial domain, a sinusoid determines an angular component of the multi-path channel between the Tx and Rx. Then, the different AoA and AoD are estimated from the parameters of their corresponding sinusoids in the mixture.

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Another methodology can be found in methods that estimate the channel in the frequency domain after applying a 2D-DFT to an initial channel estimate and estimating the DFT peaks through iterative cancellation [18], [19], [20]. Recently, the multidimensional structure of the channel has been exploited through tensor decomposition techniques to jointly estimate AoA, AoD, path delays, and Doppler shifts, achieving strong performance in scenarios with mobility and in integrated sensing and communication (ISAC) contexts [21], [22], [23], [24].

Artificial intelligence is shaping future 6G networks, and deep learning (DL) methods are being applied in wireless communications [25], including the specific case of channel estimation. In [26], a denoising neural network (NN) is considered for iterative channel estimation, while convolutional NNs (CNNs) [27], [28], Bayesian learning [29] or Transformers [30] have been applied as well to mmWave channel estimation or prediction. In the field of tracking AoA/AoD values, and assuming a single source and ABF, we can find schemes based on recurrent NNs (RNN) [31], long short-term memory (LSTM) [32] and CNN architectures [33]. Other DL methods have been proposed to decrease the amount of feedback needed for massive MIMO systems, such as [34], [35], [36].

A. Contributions

In this work, the problem of AoA and AoD estimation is addressed considering a single-source mmWave MIMO system with ABF. The choice of ABF is motivated by its cost-effectiveness and its ability to deliver high data rates in single-user scenarios [37].

In this context, two DL-based methods are proposed that exploit the sparse nature of the mmWave channel, with the aim of achieving low-complexity adaptive solutions. There is a broad literature proposing different downlink (DL) architectures that focus on sparse signal recovery for one-dimensional signals [38], [39], [40], [41] and for two-dimensional signals [42]. In our research, we approach the estimation as a supervised image-to-image translation problem. The first proposed architecture is a 2D-DeepFreq architecture, which is a 2D extension of the CNN based on a residual NN (ResNet) proposed in [38]. Since our estimation is conducted in the frequency domain rather than the transformed domain, we have omitted the matched filtering module found in the original 2D-DeepFreq design. The second proposed architecture is based on a well-known NN architecture called U-Net, which has been effectively applied to semantic segmentation [43], image super-resolution [44], and audio source separation challenges [45]. Our methodology is based on the framework proposed in [16], where the observation matrix conformed by the array outputs can be interpreted as the DFT of a sum of 2D complex sinusoids corresponding to the channel paths. This matrix is fed to the DL-based methods to achieve signal denoising and peak enhancement, providing as output a real image that encodes the underlying paths as localized Gaussian kernels. Note that the task of finding 2D frequencies is analogous to searching for local maxima in a 2D grayscale image. To enhance the output of both the ResNet and U-Net architectures, the use of a peak search algorithm based on

binary large object (blob) detection is proposed [46]. The method relies on detecting blobs, which are homogeneous or coherent regions in an image that differs in properties, i.e., brightness or color, compared their surroundings. In detection contexts, blobs are used to identify prominent local features, such as peaks, which makes the method a highly suitable approach for addressing our problem. This two-stage strategy, combining DL for super-resolution image generation with blob search for peak identification and localization, ensures a comprehensive and accurate approach to AoA/AoD estimation in ABF mmWave MIMO systems.

Our contributions are summarized as follows:

- AoA/AoD estimation in ABF mmWave MIMO is tackled using a DL-based approach. Specifically, we consider the channel matrix as a complex image and approach the solution as a super-resolution problem.
- We propose two DL-based channel estimation architectures: one based on ResNet and another based on U-Net, followed by a peak search algorithm based on blob detection.
- We conduct a comprehensive performance and complexity comparison of the proposed architectures against representative signal processing-based methods. The analysis includes robustness evaluation under hardware impairments, and performance benchmarking against the Cramér-Rao lower bound (CRLB) described in [16].

The remainder of the paper is structured as follows. Section II presents the system model and basis for the channel estimation procedure. In Section III, we describe the two DL-based AoA and AoD estimation architectures proposed in this paper. Next, in Section IV, numerical results are provided and discussed. Finally, the paper conclusions are in Section V.

Notation: Bold uppercase $\mathbf{A}$ denotes a matrix and bold lowercase $\mathbf{a}$ denotes a column vector. Superscripts T and H denote transpose and conjugate transpose, respectively. $\mathbf{I}_N$ denotes the $N \times N$ identity matrix. $\mathbf{A}_{q,p}$ is the (q, p) -th entry of matrix $\mathbf{A}$. The magnitude and phase of a complex number are denoted by $|\cdot|$ and $\angle(\cdot)$, respectively. $\mathcal{CN}(m, \sigma^2)$ is a complex Gaussian random variable with mean m and variance σ^2 . Finally, $\mathbb{C}$ and $\mathbb{R}^+$ denote the set of complex and positive real numbers, respectively.

II. SYSTEM MODEL

A. Channel Model

Let us assume a single-user narrowband mmWave MIMO geometric channel, where both the Tx and Rx are equipped with a uniform linear array of n_t and n_r antenna elements, respectively. L denotes the number of scatterers, each one contributing with a single Tx-Rx propagation path. The complex channel coefficient for each path is defined by $\alpha_l, l = 1, \dots, L$, while ψ_l and ϕ_l stand for the AoA and AoD, respectively. Using the full parametric model, the channel is expressed as [9]:

$$\mathbf{H}(\boldsymbol{\theta}) = \sqrt{n_t n_r} \sum_{l=1}^L \alpha_l \mathbf{a}_r(\psi_l) \mathbf{a}_t^H(\phi_l), \quad (1)$$

where $\boldsymbol{\theta} \triangleq [|\alpha_1|, \angle\alpha_1, \phi_1, \psi_1, \dots, |\alpha_L|, \angle\alpha_L, \phi_L, \psi_L]^T$ is the parameter vector and $|\alpha_l|$ and $\angle\alpha_l$ stand for the magnitude and phase of each channel coefficient. The α_l coefficients are independent identically distributed (i.i.d.) random variables with distribution $\alpha_l \sim \mathcal{CN}(0, \sigma_\alpha^2/L)$. AoAs (ψ_l) and AoDs (ϕ_l) are drawn from a uniform distribution $\in [0, 2\pi]$. Note that this narrowband channel model can represent the channel at each subcarrier in a wideband MIMO-OFDM system, where the channel is assumed to be frequency flat over each subcarrier.

The antenna array responses at the Tx and Rx, under a half-wavelength antenna separation assumption, can be expressed as, respectively:

$$\mathbf{a}_t(\phi_l) = \frac{1}{\sqrt{n_t}} [1, e^{-j\pi \cos \phi_l}, \dots, e^{-j\pi(n_t-1) \cos \phi_l}]^T, \quad (2)$$

$$\mathbf{a}_r(\psi_l) = \frac{1}{\sqrt{n_r}} [1, e^{-j\pi \cos \psi_l}, \dots, e^{-j\pi(n_r-1) \cos \psi_l}]^T. \quad (3)$$

The problem of estimating the channel $\mathbf{H}(\boldsymbol{\theta})$ becomes a problem of estimating the parameters in $\boldsymbol{\theta}$. As already mentioned, mmWave channels are highly sparse, as demonstrated by measurements [11]. Hence, the L paths are likely to be separated from each other, which simplifies the channel estimation task.

B. Pilot-Based Training Phase

The first step is the open-loop pilot-based training phase. The beam search space is given by a codebook comprising P and Q codewords/directions at the Tx and Rx side, respectively. The observation matrix is formed after transmitting the pilot symbol through the $Q \times P$ direction combinations.

The process is as follows: A pilot symbol s known by Tx and Rx is transmitted and received through a subset of P and Q codewords or directions at the Tx and Rx side, respectively, leading to quantized angles $\bar{\phi}_p$, $p = 0, 1, \dots, P-1$ and $\bar{\psi}_q$, $q = 0, 1, \dots, Q-1$. Due to the analog beamforming scenario, the Tx and Rx have only one radio-frequency chain, so the beamforming and combining operations are carried out in the analog domain [16].

Beamforming/combining vectors are calculated to match the channel response [47], so $\mathbf{f} = \mathbf{a}_t(\bar{\phi}_p)$ for $p = 0, 1, \dots, P-1$, and $\mathbf{w} = \mathbf{a}_r(\bar{\psi}_q)$ for $q = 0, 1, \dots, Q-1$. For each $\{q, p\}$ direction pair, the received signal is

$$y_{q,p} = \sqrt{\rho} \mathbf{w}_q^H \mathbf{H} \mathbf{f}_p s + \mathbf{w}_q^H \mathbf{n}, \quad (4)$$

where $\rho \in \mathbb{R}^+$ is the transmit power. The noise term $\mathbf{n} \sim \mathcal{CN}(0, \Sigma_{\mathbf{n}})$ is a complex additive white Gaussian noise $1 \times n_r$ vector with covariance $\Sigma_{\mathbf{n}} = \sigma_n^2 \mathbf{I}_{n_r}$, where $\mathbf{I}_{n_r}$ stands for the $n_r \times n_r$ identity matrix. The symbol s is set to 1 during training for the sake of simplicity. Hence, the system signal-to-noise ratio is given by ρ/σ_n^2 .

The following observation matrix $\mathbf{Y} \in \mathbb{C}^{Q \times P}$ is obtained after transmitting the pilot symbols through the $Q \times P$ directions

$$\mathbf{Y} = \begin{bmatrix} y_{0,0} & y_{0,1} & \cdots & y_{0,P-1} \\ y_{1,0} & y_{1,1} & \cdots & y_{1,P-1} \\ \vdots & \vdots & \ddots & \vdots \\ y_{Q-1,0} & y_{Q-1,1} & \cdots & y_{Q-1,P-1} \end{bmatrix} = \sqrt{\rho} \mathbf{G}(\boldsymbol{\theta}) + \mathbf{N}. \quad (5)$$

The noise matrix $\mathbf{N} \in \mathbb{C}^{Q \times P}$ contains i.i.d. $\sim \mathcal{CN}(0, \sigma_n^2)$ elements, and $\mathbf{G} \in \mathbb{C}^{Q \times P}$ encodes the channel parameter vector $\boldsymbol{\theta}$.

Elements in the observation matrix can be rearranged to separate the effect of the different path components

$$\mathbf{Y} = \sqrt{\rho} \sum_{l=1}^L \mathbf{G}^{(l)}(\boldsymbol{\theta}_l) + \mathbf{N}. \quad (6)$$

In the above expression, the observation matrix is written as a sum of path contributions $\mathbf{G}^{(l)}(\boldsymbol{\theta}_l) \in \mathbb{C}^{Q \times P}$, each one being dependent on a parameter vector $\boldsymbol{\theta}_l = [|\alpha_l|, \angle\alpha_l, \phi_l, \psi_l]^T$.

Since the beamforming/combining vectors are designed to match the array response, the elements $g_{q,p}^{(l)} = [\mathbf{G}^{(l)}(\boldsymbol{\theta}_l)]_{q,p}$ are given by

$$g_{q,p}^{(l)}(\boldsymbol{\theta}_l) = A_l \frac{1 - e^{-j\pi n_r(\cos \psi_l - \cos \bar{\psi}_q)}}{1 - e^{-j\pi(\cos \psi_l - \cos \bar{\psi}_q)}} \frac{1 - e^{j\pi n_t(\cos \phi_l - \cos \bar{\phi}_p)}}{1 - e^{j\pi(\cos \phi_l - \cos \bar{\phi}_p)}}, \quad (7)$$

where $A_l = \alpha_l / \sqrt{n_t n_r}$.

Considering a specific DFT-based codebook for ABF, the observation matrix $\mathbf{Y}$ corresponds to the 2D-DFT of a sum of windowed complex sinusoids embedded in AWGN. Note that the elements of path components $\mathbf{G}^{(l)}(\boldsymbol{\theta}_l)$ in (7) can be equivalently expressed as

$$g_{q,p}^{(l)}(\boldsymbol{\omega}_l) = A_l \frac{1 - e^{-j(\omega_q - \omega_{\psi_l})n_r}}{1 - e^{-j(\omega_q - \omega_{\psi_l})}} \frac{1 - e^{-j(\omega_p - \omega_{\phi_l})n_t}}{1 - e^{-j(\omega_p - \omega_{\phi_l})}}, \quad (8)$$

where

$$\omega_q = -\pi \cos(\bar{\psi}_q), \quad \omega_p = \pi \cos(\bar{\phi}_p), \quad (9)$$

$$\omega_{\psi_l} = -\pi \cos(\psi_l), \quad \omega_{\phi_l} = \pi \cos(\phi_l). \quad (10)$$

As demonstrated in [16], $g_{q,p}^{(l)}$ is the (q, p) coefficient of the 2D-DFT of $A_l e^{j(\omega_{\psi_l} m + \omega_{\phi_l} n)}$. The variables $\omega_{\psi_l}, \omega_{\phi_l} \in [-\pi, \pi]$ denote the frequencies of such complex sinusoid in each spatial direction, where the vertical direction is related to the AoA and the horizontal direction to the AoD. Note that the dependency on $\boldsymbol{\theta}_l$ in (7) has been changed to $\boldsymbol{\omega}_l = [|\alpha_l|, \angle\alpha_l, \omega_{\phi_l}, \omega_{\psi_l}]^T$ in (8) to emphasize the focus on the spatial frequencies rather than on the AoDs and AoAs.

III. DEEP-LEARNING-BASED AOA AND AOD ESTIMATION

In this work, two DL architectures are proposed to address the AoA and AoD estimation problem as a search for the spectral peaks contained in the 2D image representing the values in the observation matrix $\mathbf{Y}$. More precisely, the estimation is tackled as a supervised image-to-image translation problem, where images from one domain are transformed so they have the style (or characteristics) of images from another domain.

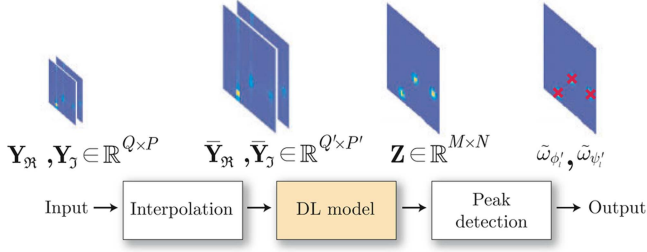


Fig. 1. Main stages of the proposed AoAs and AoDs estimation.

A. Problem Formulation

Fig. 1 summarizes the steps followed to obtain the AoA and AoD estimates. In a preprocessing stage, the complex-valued observation matrix $\mathbf{Y}$ is decomposed into its real and imaginary parts $\mathbf{Y}_{\mathcal{R}} \in \mathbb{R}^{Q \times P}$ and $\mathbf{Y}_{\mathcal{I}} \in \mathbb{R}^{Q \times P}$, respectively, following the proposal in [42]. The latter matrices are modified through an interpolation stage which increases their dimensions to $Q' = \beta Q$ and $P' = \beta P$, resulting in new matrices $\bar{\mathbf{Y}}_{\mathcal{R}} \in \mathbb{R}^{Q' \times P'}$ and $\bar{\mathbf{Y}}_{\mathcal{I}} \in \mathbb{R}^{Q' \times P'}$.

$$\begin{aligned}\bar{\mathbf{Y}}_{\mathcal{R}} &= \mathcal{I}(\mathbf{Y}_{\mathcal{R}}), \\ \bar{\mathbf{Y}}_{\mathcal{I}} &= \mathcal{I}(\mathbf{Y}_{\mathcal{I}}),\end{aligned}$$

where $\mathcal{I}(\cdot)$ denotes a function which interpolates the input through the well-known nearest-neighbor method [48], and $\beta \in \mathbb{Z}_+$ is a design parameter. In this work, interpolation with $\beta \in \{2, 4\}$ has been applied to set the size of the input signal equal to 64×64 . In the problem tackled by the proposed DL-based approaches, the domain corresponds to the set of matrices $\bar{\mathbf{Y}}_{\mathcal{R}}$ and $\bar{\mathbf{Y}}_{\mathcal{I}}$ from which we seek to estimate the set of AoAs and AoDs. The input of the network is, thus, of size $Q' \times P' \times 2$, whereas the network gives as an output a matrix $\mathbf{Z} \in \mathbb{R}^{M \times N}$. The parameters M and N are selected as integer multiples of the input dimensions Q' and P' .

The output of the DL architectures is interpreted as a denoised observation matrix containing a set of local maxima that identifies the mmWave channel paths. Thus, a peak search algorithm is used to find the corresponding AoAs and AoDs in the test stage. Note that the task of finding 2D frequencies is analogous to searching for local maxima in a 2D grayscale image. A peak search algorithm based on blob detection principles is implemented, which is well-suited for identifying bright, compact regions within an intensity map.

In the following, we first detail the DL architectures and their training configuration. Then, we describe the basics of the peak search algorithm.

B. NN 1: ResNet Architecture

The first approach involves the utilization of a 2D-DeepFreq architecture, a modification of the ResNet-based CNN proposed in [42] (see Fig. 2). A preliminary version of this scheme was presented in [49]. This adapted ResNet incorporates **transposed convolution layers** to achieve super-resolution. The method addresses the AoA and AoD estimation problem by identifying

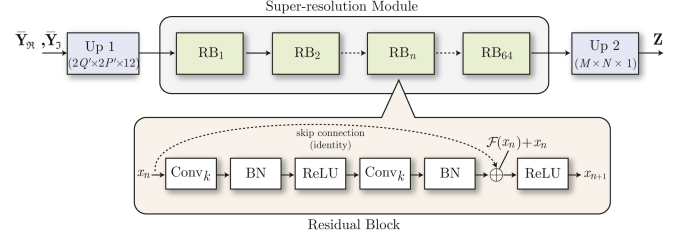


Fig. 2. ResNet architecture. Output dimensions are specified in parentheses.

the complex sinusoids present in the input complex-valued observation matrix $\mathbf{Y}$. In contrast to the 2D-DeepFreq model, our input $\mathbf{Y}$ already encodes a frequency-domain representation of the sum of multiple sinusoids. Hence, the match filtering module present in DeepFreq before the upsampling stage has been suppressed. As a fully convolutional model, the architecture is based on convolutional layers, denoted as Conv_k , where k indicates the length of their corresponding (square) $k \times k$ filter kernels. We denote by w_n , h_n , and c_n the width, height and number of channels of the output feature map at the n -th layer, respectively, while f_n stands for the number of filters. The number and size of all filters utilized in the convolutions is consistently set to $f_n = 12$ and $k_n = k = 5$, preserving the size of the feature maps with zero-padding, as specified in [42].

As shown in Fig. 2, the input is formed by the interpolated real and imaginary components of the observation matrix, i.e. $\bar{\mathbf{Y}}_{\mathcal{R}}$ and $\bar{\mathbf{Y}}_{\mathcal{I}}$, stacked in the channels dimension. These enter an upsampling stage (Up 1) using a transposed 2D convolutional layer (ConvT_k) with a stride of 2, leading to an output of dimensions $2Q' \times 2P' \times 12$.

Then, the image is processed by the super-resolution module which iteratively stacks 64 residual blocks (RBs), as described in [45]. As shown in Fig. 2, RBs follow the Conv_k -BN-ReLU order, where ReLU is the rectified linear unit function and BN stands for batch normalization. For a generic input tensor x to any RB, the output is given by:

$$\text{RB}(x) := \text{ReLU}(\mathcal{F}(x) + x),$$

where

$$\mathcal{F}(x) := \text{BN}(\text{Conv}_k(\text{ReLU}(\text{BN}(\text{Conv}_k(x)))).$$

Finally, another transposed convolutional layer ConvT_k with a stride of 2 is applied for a second upsampling (Up 2). Therefore, the output of the network is a matrix $\mathbf{Z} \in \mathbb{R}^{M \times N}$.

C. NN 2: U-Net Architecture

The second proposed approach is grounded in the well-established U-Net architecture, initially developed for biomedical image segmentation tasks [43]. The U-Net constitutes a fully CNN designed for tasks involving substantial data volumes and high variability. The architecture, named for its characteristic “U” shape, comprises a contracting path for context capture and a symmetric expanding path facilitating precise localization. In this study, we adopt the Wave-U-Net variant, introduced for audio source separation [45].

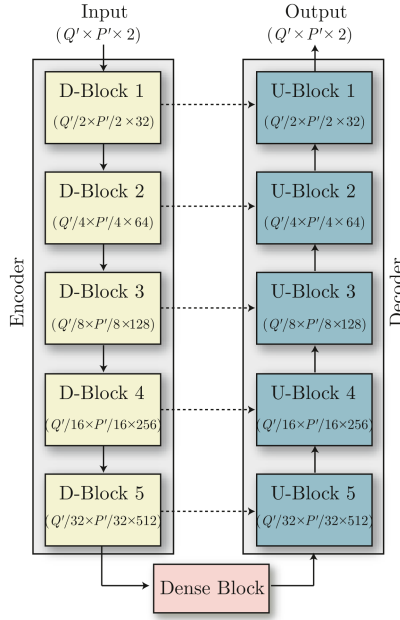


Fig. 3. U-Net Block. Output dimensions are specified in parentheses.

The main structure within the U-Net model is the U-Net Block, which is detailed in Fig. 3. The U-Net Block has an encoder part (left) and a decoder part (right), made up of depth blocks (D-Block) and upsampling blocks (U-Block), respectively. The encoder comprises five D-Blocks defined as

$$\text{D-Block}(x) := \text{MaxPool}(\text{ConvBlock}(\text{ConvBlock}(x))),$$

where x is the input tensor, MaxPool denotes a max-pooling layer and ConvBlock stands for a standard convolutional block (with $k = 5$), as follows:

$$\text{ConvBlock}(x) = \text{ReLU}(\text{BN}(\text{Conv}_k(x))),$$

Between the encoder and decoder of the U-Net Block, there is a Dense Block. This Dense Block flattens an input tensor x of shape (B, H, W, C) into a 2D tensor of shape $(B, H \times W \times C)$ through a flattening operator (Fl), passing the output through two dense layers and a final reshaping operator (Rshp) to match the decoder input:

$$\text{DenseBlock}(x) = \text{ConvBlock}(\text{Rshp}(\text{Dense}_2(\text{Dense}_1(\text{Fl}(x))))))$$

While the number of neurons in Dense₂ should match the required size for the decoder input, Dense₁ establishes the bottleneck size, which we selected to be 1024. Similarly, the decoder comprises five U-Blocks and a super-resolution layer. Each U-Block consists of a ConvT_k layer, BN and ReLU activation, as follows:

$$\text{U-Block}_i(x) = \text{ConvBlock}(\text{ConvBlock}(\text{ConvT}_k(x \oplus y_i))),$$

where y_i denotes the output of the corresponding D-Block at the encoder and $\oplus$ means concatenation. Finally, both the input and the output of the U-Net block go, respectively, through the upsampling layers “Up 0” and “Up 1” before being concatenated and passed to the last “Up 2” layer, as shown in Fig. 4.

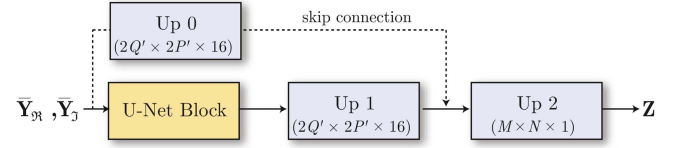


Fig. 4. U-Net overall architecture.

D. Training Configuration

As in [42], during the training stage, the considered ground truth for each of the channels in the dataset corresponds to a high-resolution matrix $\mathbf{X} \in \mathbb{R}^{M \times N}$ built as the sum of L 2D Gaussian functions. The elements of matrix $\mathbf{X}$, for $m = 0, \dots, M - 1$ and $n = 0, \dots, N - 1$, are constructed as follows

$$\mathbf{X}_{m,n} = \frac{1}{2\pi\sigma_1\sigma_2} \sum_{l=1}^L e^{\left(\frac{(\omega_m - \tilde{\omega}_{\psi_l})^2}{2\sigma_1^2} + \frac{(\omega_n - \tilde{\omega}_{\phi_l})^2}{2\sigma_2^2} \right)}, \quad (11)$$

where $\omega_m = m \frac{2\pi}{M}$, $\omega_n = n \frac{2\pi}{N}$, and $\tilde{\omega}_{\phi_l} = \mathcal{W}_{2\pi}(\omega_{\phi_l})$, $\tilde{\omega}_{\psi_l} = \mathcal{W}_{2\pi}(\omega_{\psi_l})$ stand for the 2π -wrapped ground-truth frequencies. The parameters σ_1 and σ_2 represent the standard deviations that define the width of the Gaussian functions. We consider the values $\sigma_1 = \sigma_2 = 0.07$. The models are trained using the mean square error (MSE) between the ground truth and the NN output as loss function:

$$\text{Loss} = \frac{1}{MN} \sum_{n=0}^{N-1} \sum_{m=0}^{M-1} (\mathbf{Z}_{m,n} - \mathbf{X}_{m,n})^2.$$

E. Peak Detection Method

The peak detection method is based on detecting bright, spatially compact regions (referred to as blobs) through four main stages [46]. It is tailored to exploit the structure of the denoised observation matrices produced by the deep learning architectures, which exhibit localized energy concentrations corresponding to dominant channel paths in the angular domain, analogous to bright peaks in a two-dimensional grayscale image.

- 1) *Preprocessing*: The predicted observation matrix is first normalized to an 8-bit grayscale representation. This step ensures compatibility with intensity-based detection techniques and allows for consistent treatment across varying signal magnitudes. Optionally, a binary thresholding operation may be applied to eliminate low-energy regions that may correspond to noise or diffuse responses. Pixels with intensity values below a fixed threshold are suppressed, while those above it are saturated to emphasize potential peaks.
- 2) *Blob Detection*: The core detection mechanism identifies blobs within the preprocessed matrix. The detector applies constraints based on the intensity of the pixels and the area of the blob, retaining only the regions that fall within a specified size range. Crucially, no geometric assumptions (e.g., circularity or convexity) are imposed, allowing for the detection of irregularly shaped or distorted peaks that may arise from propagation or estimation effects.

Algorithm 1: 2D-NN for AoA and AoD Estimation.**Model Training****Input:** Complex data samples $(\mathbf{Y}_{\mathbb{R}}, \mathbf{Y}_{\mathbb{I}})$, ground truth path matrix $\mathbf{X}$ **Generate Dataset:** $\mathcal{D} \leftarrow \{(\mathbf{Y}_{\mathbb{R}}, \mathbf{Y}_{\mathbb{I}}), \mathbf{X}\}_{i=1}^{N_{data}}$ **Train NeuralNet:** $\text{NeuralNet} \leftarrow \text{TrainNeuralNet}(\mathcal{D})$ **Result:** Trained model NeuralNet **Angle Recovery****Predict:** $\mathbf{Z} \leftarrow \text{NeuralNet}(\mathbf{Y}_{\mathbb{R}}, \mathbf{Y}_{\mathbb{I}})$ **Output:** Angles $\{(\tilde{\omega}_{\phi_l}, \tilde{\omega}_{\psi_l})\}_{l=1}^L \leftarrow \text{PeakDetection}(\mathbf{Z})$

- 3) **Peak Localization:** Each detected blob is reduced to a key-point representing its centroid, interpreted as the angular coordinates of a candidate path (AoA/AoD). These coordinates are computed with subpixel precision, enabling accurate estimation of the underlying 2D frequency components. The method inherently supports multiple peak detections, making it suitable for multipath scenarios.
- 4) **Peak Ranking:** To prioritize the most significant propagation paths, the detected peaks are re-evaluated using the original (non-thresholded) intensity map. Each peak is assigned an amplitude score corresponding to the local value of the matrix at the detected location. The peaks are then sorted in descending order of amplitude, ensuring that the most dominant paths are selected for subsequent processing.

This blob-based peak detection framework provides a robust and efficient means of identifying the spatial signatures of mmWave channel components. Its resilience to noise, flexibility with respect to shape, and capacity to capture multiple high-resolution components make it well aligned with the structure of the learned outputs and the demands of angular domain channel estimation.

Algorithm 1 summarizes the main steps followed by the proposed NN architectures.

IV. NUMERICAL RESULTS

In this section, we assess the performance of the proposed ResNet and U-Net architectures through numerical experiments. The methods are compared in terms of performance and complexity with the TSDCE and DFT-based channel estimation algorithm (DFT-CEA) channel estimation methods. The performance is also compared to the CRLB of the 2D-frequency estimation problem. The AoA/AoD estimation accuracy is assessed through the root mean square error (RMSE) of the angle measurements, which is calculated as

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{N}|} \sum_{i \in \mathcal{N}} (\vartheta_i - \hat{\vartheta}_i)^2}, \quad (12)$$

where $\mathcal{N}$ denotes the set of successfully detected angles and ϑ_i considers any AoA or AoD. Note that a detection is considered successful if the associated $\text{RMSE} \leq 1^\circ$. A second metric jointly evaluated is the probability of successful detection (P_D) at different signal-to-noise ratios (SNRs).

A. Baselines for Comparison

To assess the performance of the proposed approaches, we consider three benchmark schemes: TSDCE, DFT-CEA and CRLB. These benchmarks were chosen based on prior work demonstrating that TSDCE outperformed other commonly used methods, such as LS and OMP, while DFT-CEA showed superior performance compared to TSDCE under specific conditions. Additionally, the CRLB serves as a useful theoretical lower bound.

The benchmarks are as follows:

- 1) TSDCE method. It is an iterative algorithm which starts by taking the observation matrix $\mathbf{Y}$ and applies a 2D inverse FFT to it, i.e., it operates in the transformed spatial domain. Next, it uses a cropping procedure to extract the important information from the matrix, creating a new matrix called $\bar{\mathbf{D}}'_C$. The SVD of this new matrix is next used to estimate the contribution of the largest gain path component. Afterwards, the algorithm uses the 2D-autocorrelation function (ACF) to reduce the noise level and to obtain the phase angles, which are used to estimate the ψ_l and ϕ_l parameters. The rest of the paths are estimated using the same procedure, although the cropped matrix is updated by successive interference cancellation to remove the contribution of the estimated paths.
- 2) DFT-CEA. This method works by obtaining an initial channel estimate and applying a 2D-DFT of N_{DFT} points to it. Then, it analyzes the peaks of the DFT output in order to estimate the AoAs and AoDs of the main channel paths. This process is done repeatedly until the angles are accurately estimated. As in [18], we set $N_{DFT} = 1024$ points for the DFT-CEA method.
- 3) CRLB for channel estimation methods based on two-dimensional frequency estimation. In [16], the CRLBs of the variance for the estimate of each channel parameter were used to obtain a lower bound performance. This bound was useful to assess channel estimation methods based on two-dimensional frequency processing such as TSDCE. In this work, we will also include this performance bound within the performance evaluation.

B. ResNet and U-Net Parameters

For the training and validation of the ResNet and U-Net architectures, datasets composed of 10000 and 1000 different observation matrices are used, respectively. The datasets were synthetically generated following the model in (5) with $\rho = 1$, i.e. the SNR definition is $1/\sigma_n^2$. While the validation dataset remains fixed throughout the optimization process, the training examples vary with each epoch, effectively resulting in an infinite training dataset.¹ More specifically, each dataset element contains a random observation matrix generated for a random SNR in the range between -10 dB and 25 dB, for a number of channel paths randomly selected from $L = 1$ to 10 . The codebook size is also selected randomly by setting $Q = P$ to either 16 or 32 . Channel coefficients $\alpha_l, l = 1, \dots, L$, are drawn

¹Code available at <https://github.com/SandraRoger/DLDOA>

TABLE I
TRAINING PARAMETERS FOR THE RESNET AND U-NET

Parameter	ResNet	U-Net
Optimizer	RMSprop	Adam
Learning Rate	0.003	0.001
Batch size	32	32
Epochs	500	500
Training data size	10000	10000
Validation data size	1000	1000

from a zero mean complex Gaussian distribution with variance $1/L$, while AoA and AoD angles are drawn from a uniform random distribution in the range $[0, \pi]$. Each dataset element also contains the corresponding ground-truth target image, with 2D Gaussian functions centered at the corresponding AoA and AoD for the L channel paths. The architectural hyperparameters of both networks, such as the number of layers, filter sizes, and activation functions, were selected using the KerasTuner library. This framework allowed us to perform an automated hyperparameter search over a wide configuration space. The architectures reported in this work correspond to the configurations that consistently achieved the best validation performance across a range of SNR values. Similarly, the target Gaussian width ($\sigma_1 = \sigma_2 = 0.07$) was selected as the best performing value considering $\sigma_1, \sigma_2 \in [0.2, 0.1, 0.07, 0.05]$. Although a detailed ablation study is out of scope, analyzing the sensitivity of performance to architecture variations remains a relevant direction for future research. Table I shows other considered training parameters for both NNs.

The simulations carried out for the performance evaluation consider 1000 random channel realizations for each SNR value in the displayed range, assuming the same distribution of α_l and AoA and AoD as for training. For a fair comparison, the same pilot overhead is used for evaluating all the algorithms, which is directly $Q \times P$ pilots. Also, the number of desired paths is set to L and a single radio frequency (RF) chain is assumed for all methods.

C. Performance Evaluation

Figs. 5 and 6 show the P_D and RMSE of the estimated AoA/AoD in channels with $L = 3$ obtained with the ResNet and U-Net approaches, respectively, considering $n_T = n_R = 16$ and a codebook with $Q = P = 16$. For each NN type, two models with different output sizes have been tested with the same architecture but different upsampling factor in the last "Up 2" layer, leading to $M = N \in \{256, 512\}$. Recall that the displayed RMSE values consider only the set of successfully detected angles $\mathcal{N}$, which depends on P_D and, thus, it differs among algorithms.

It can be observed in Fig. 5 that the ResNet architecture achieves similar results both in P_D and in RMSE for the two

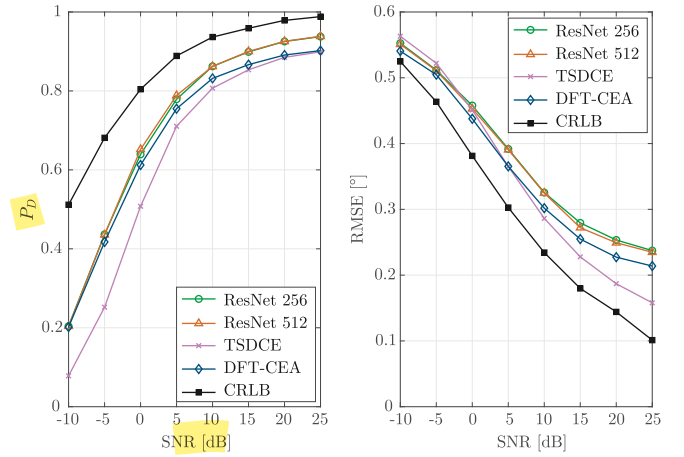


Fig. 5. P_D and RMSE of AoA/AoD estimation for the ResNet model with $L = 3$, $n_t = n_r = 16$ and $Q = P = 16$.

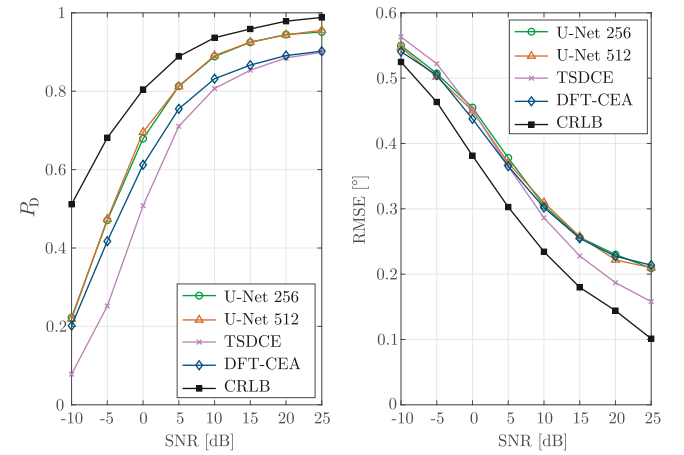


Fig. 6. P_D and RMSE of AoA/AoD estimation for the U-Net model with $L = 3$, $n_t = n_r = 16$ and $Q = P = 16$.

evaluated values of M , with only a slight advantage for ResNet 512 at medium SNR levels. This is an important finding since it implies that opting for $M = 256$ provides comparable accuracy with reduced computational complexity. Considering the P_D metric, ResNet approaches are superior to signal processing methods, establishing themselves as the method to maximize this parameter especially at high SNRs. In the RMSE metric, ResNet models are competitive for low-to-medium SNRs, while signal processing-based methods such as TSDCE show improved performance for high SNR. Nevertheless, the ResNet performance remains close to that of DFT-CEA across the full SNR range.

Fig. 6 provides a comparative analysis of U-Net for both $M = N$ values. Similar to ResNet, the U-Net architecture exhibits nearly indistinguishable performance for both M values, which is advantageous in terms of computational complexity. With respect to the P_D metric, U-Net outperforms the signal processing-based methods across the entire SNR range, narrowing the gap to the CRLB at high SNR. Regarding RMSE, U-Net matches the performance of DFT-CEA and clearly surpasses TSDCE at low and medium SNRs. Although TSDCE continues

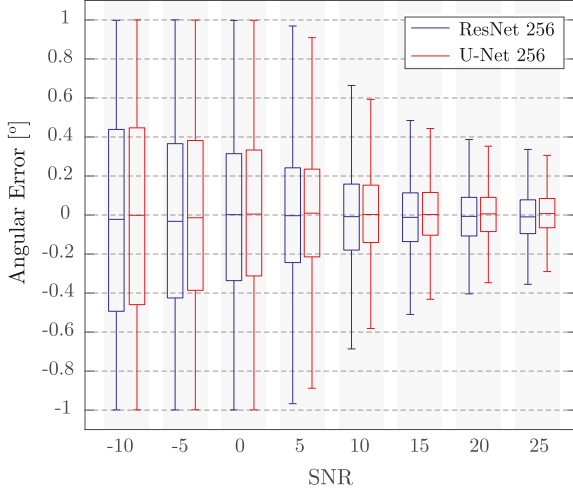


Fig. 7. Boxplots of the AoA/AoD estimation errors obtained by the ResNet and U-Net with $M = 256$ for different SNR values, with $L = 3$, $n_t = n_r = 16$ and $Q = P = 16$.

to reduce RMSE at high SNR, it does so at the expense of having the lowest P_D among all evaluated methods.

The joint analysis from Figs. 5 and 6 results reveal a favorable trade-off between P_D and RMSE for both ResNet and U-Net. The proposed NNs architectures show strong detection capabilities with a competitive estimation accuracy. The reason behind the slightly larger RMSE values is that these NN-based methods are capable of detecting a wider set of AoA/AoD angles, including those near the critical 1° threshold, which, while slightly increasing RMSE due to a larger averaging set, reflects a higher detection sensitivity. This behavior underlines the versatility of the proposed schemes, which is very valuable in real-world applications where wireless communication faces challenging signal environments, ranging from weak and noisy conditions to more favorable ones.

To gain further insight into the differences between the ResNet and U-Net, Fig. 7 presents boxplots of the angle estimation errors. This comparison includes only the cases with $M = N = 256$ for different values of SNR, with $L = 3$, $n_t = n_r = 16$ and $Q = P = 16$. The plotted angular errors correspond to the individual estimates used to compute the RMSE values shown in previous figures, but here they are visualized to assess the spread and distribution of errors. Overall, the U-Net shows a better fit to the median, and both models display symmetric error distributions, indicating no apparent bias. At low SNRs (from -10 to 0 dB), according to the upper and lower quartiles and extreme values, both schemes exhibit a wide range of angle values. The situation starts to change at $SNR = 5$ dB, where the U-Net presents more reduced extreme values than the ResNet for medium to high SNRs (between 10 and 25 dB).

The effect of increasing the number of paths and the robustness of the evaluated methods can be seen in Fig. 8, which shows the results of varying the parameter L for two SNR values, setting $M = N = 256$ in the NNs. Results in P_D for both SNR values show that the ResNet and U-Net are able to detect more angles than the baselines as the number of paths increases, implying a larger robustness against the L parameter

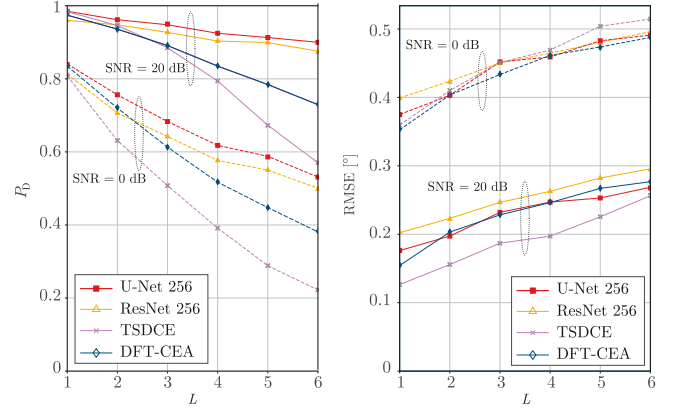


Fig. 8. P_D and RMSE of AoA/AoD estimation errors for different values of L , $n_t = n_r = 16$ and $Q = P = 16$ at two representative SNR values: 0 dB (dashed lines) and 20 dB (solid lines).

than the signal processing-based methods. Indeed, the P_D curves of DL-based schemes present a smaller slope than the one of the DFT-CEA and TSDCE methods, specially for high SNR. Comparing U-Net and ResNet, we can observe that the U-Net achieves better results, remaining almost a constant gap between both architectures as the parameter L increases. In terms of RMSE, signal-processing-based methods benefit from detecting a smaller set of angles, so DFT-CEA achieves the best results for low SNR, while TSDCE is the one providing the most accurate angle estimation for the considered range of L . In this case, the NNs manage to remain close to DFT-CEA in low SNR and $L > 3$ values, while only the U-Net is able to outperform the DFT-CEA performance at high SNR.

D. Performance Under Hardware Impairments

In practical ABF systems, each antenna element applies a specific phase shift to steer the beam in a desired direction. However, hardware imperfections in the phase shifters can introduce small deviations from the intended phase values, thereby distorting the overall beam pattern. To evaluate the influence of such imperfections, we model the phase shift error associated with the i -th antenna, denoted by δ_i , as a random variable uniformly distributed over the range $[-\delta_{\max}, \delta_{\max}]$, with $\delta_{\max} \in (0^\circ, 180^\circ]$. This representation captures the variability introduced by non-ideal hardware implementations [50].

Assuming independent errors across antennas, the effective phase shift applied to the i -th element of the Tx/Rx antenna array is perturbed by δ_i , thus altering the array responses in Eqs. (2)–(3). The experimental evaluation in this section quantifies the degradation in angular estimation performance when such phase deviations are present. In our evaluation, we consider three levels of phase-shifter error with values $\delta_{\max} = 1^\circ, 2^\circ$, and 5° . These values are selected to represent realistic hardware impairments, with $\delta_{\max} = 1^\circ$ corresponding to high-quality implementations, and $\delta_{\max} = 5^\circ$ included as a pessimistic scenario to assess the robustness of the proposed methods under severe hardware impairments, as shown in [51].

TABLE II
PERFORMANCE AT SNR = 0 dB AND 20 dB ACROSS METHODS AND PHASE ERROR LEVELS ($\delta_{\max}$)

$\delta_{\max}$	SNR = 0 Db								SNR = 20 Db							
	ResNet		U-Net		TSDCE		DFT-CEA		ResNet		U-Net		TSDCE		DFT-CEA	
	P_D	RMSE	P_D	RMSE	P_D	RMSE	P_D	RMSE	P_D	RMSE	P_D	RMSE	P_D	RMSE	P_D	RMSE
0°	0.643	0.458	0.679	0.454	0.524	0.446	0.622	0.437	0.925	0.253	0.945	0.230	0.885	0.184	0.891	0.227
1°	0.643	0.458	0.678	0.454	0.524	0.446	0.622	0.437	0.923	0.253	0.945	0.231	0.885	0.187	0.888	0.234
2°	0.642	0.458	0.677	0.454	0.518	0.447	0.622	0.438	0.921	0.254	0.941	0.234	0.883	0.190	0.888	0.235
5°	0.638	0.461	0.677	0.456	0.508	0.449	0.621	0.438	0.909	0.279	0.925	0.255	0.880	0.215	0.884	0.239

TABLE III
COMPUTATIONAL COMPLEXITY OF THE 2D RESNET MODEL

Block	Layer	Computational complexity
Up1	1	$2^3 5^2 3 P' Q'$
Super-resolution	2 to 65	$64(2^8 5^2 3 P' Q')$
Up2	66	$2^7 5^2 P' Q'$

Table II presents the performance of all methods under different phase error levels at SNRs of 0 dB and 20 dB. At 0 dB, the learning-based approaches (ResNet and U-Net) achieve significantly higher detection probabilities and lower RMSE compared to traditional signal processing methods, even when phase errors are introduced. The effect of hardware impairments is less pronounced in this low-SNR regime, likely because noise remains the dominant source of error. At 20 dB, where impairments have a more visible impact, the learning-based models still outperform the classical approaches across all levels of distortion. The DFT-CEA exhibits a more stable behavior at high SNR than the other methods, however, its accuracy and detection performance remain below those of the deep learning methods, reinforcing the advantage of data-driven architectures in both ideal and non-ideal scenarios. One promising direction to further enhance robustness of the NN models to hardware impairments is to retrain the current models using additional training data containing impairments.

E. Complexity Analysis

The computational complexity of the ResNet CNN architecture can be obtained by analyzing its different stages. As shown in [42], the complexity of the n -th convolutional layer can be calculated as $h_n w_n c_{n-1} k_n^2 f_n$ and the complexity of the n -th transpose convolutional layer is $h_{n-1} w_{n-1} c_{n-1} k_n^2 f_n$. Considering the parameters of the ResNet layers specified in Section III-B, Table III summarizes the resulting computational complexity for each one of the blocks with respect to the size of the input signal ($\bar{Y}_R$ and $\bar{Y}_3$). After observing the complexity of all the convolutional layers and performing an asymptotic analysis, it can be seen that the method involves an overall complexity order $\mathcal{O}(P'Q')$, thus, $\mathcal{O}(PQ)$, since $P' = \beta P$ and $Q' = \beta Q$, with $\beta \in \{2, 4\}$.

In the case of the U-Net, we perform a complexity analysis following the methodology in [42]. Table IV summarizes the computational complexity for each block and layer depending on the size of the input signal. Regarding the asymptotical analysis and following the same reasoning as for the ResNet, the complexity related to the encoder and decoder is then in the order $\mathcal{O}(PQ)$. In the case of the dense block, asymptotically,

TABLE IV
COMPUTATIONAL COMPLEXITY OF THE 2D U-NET MODEL

Block	Layer	Computational complexity
Enc	1 and 2	$2^6 3^2 P' Q' + 2^{10} 3^2 P' Q'$
Enc	3 to 10	$4(2^9 3^2 P' Q' + 2^{10} 3^2 P' Q')$
Dense	1 and 2	$2^{-2}(P' Q')^2 + 2^{-7}(P' Q')^2$
Dec	1 and 2	$2^5 3^2 P' Q' + 2^{10} 3^2 P' Q' + 2^9 3^2 P' Q'$
Dec	3 to 15	$5(2^{11} 3^2 P' Q' + 2^{10} 3^2 P' Q' + 2^9 3^2 P' Q')$
Dec	16 to 19	$2^5 3^2 P' Q' + 2^{16} 3^2 P' Q' + 2^{11} 3^2 P' Q'$

the overall complexity is also of order $\mathcal{O}(PQ)$. Hence, before the peak detection stage, both NN architectures show an asymptotical complexity that is dominated by the term $\mathcal{O}(PQ)$.

The peak detection stage is implemented using the `cv::SimpleBlobDetector` function from OpenCV [52]. This detector is configured for detecting bright blobs in an image. In our implementation, shape-based filters like circularity, convexity, and inertia are ignored, and the threshold range for blob detection is set from 0 to 255. The complexity of the peak detection stage depends on the size of the input image, which in our case is proportional to $P' = \beta P$ and $Q' = \beta Q$, and thus, to P and Q , since $\beta \in \{2, 4\}$. Following the stages of the blob detector in subsection III-E, the main complexity arises from the blob detection ($\mathcal{O}(PQ)$ due to pixel scanning) and peak ranging ($\mathcal{O}(L \cdot \log L)$). Assuming that, in a mmWave channel, $L < \{P, Q\}$, the complexity of the peak detection scheme is dominated as well by the term $\mathcal{O}(PQ)$ as in the NN processing stage, and an overall complexity of $\mathcal{O}(PQ)$ can be assumed for the NN architectures.

The algorithms used as benchmark in this section show an asymptotic complexity of $\mathcal{O}((L-1) \cdot PQ \min(P, Q))$ and $\mathcal{O}(N_{DFT}^2 \log_2 N_{DFT})$, for the TSDCE and DFT-CEA [16], respectively, while the NNs under consideration present a complexity dominated by $\mathcal{O}(PQ)$. Note that, in general, $N_{DFT} > \{P, Q\}$, so that DFT-CEA is the most asymptotically expensive algorithm. Interestingly, NNs exhibit a competitive asymptotical complexity also with respect to the TSDCE, specially for large values of $\{P, Q\}$ as the ones in massive MIMO, due to the multiplicative $\min(P, Q)$ factor in the TSDCE complexity order.

V. CONCLUSION

In this work, we proposed and evaluated two deep-learning-based approaches estimating the angles-of-arrival (AoAs) and angles-of-departure (AoDs) in a parametric mmWave MIMO channel, specifically, through adapting and extending both a residual convolutional neural network (ResNet) and a U-Net. The estimation problem was framed as a supervised image-to-image translation task in the frequency domain.

Our numerical simulations showed that both proposed architectures achieve competitive performance compared to traditional signal processing techniques, with the U-Net consistently outperforming all evaluated methods in terms of detection probability (P_D) across a wide SNR range. In terms of estimation accuracy (RMSE), while signal processing methods such as transformed spatial domain channel estimation (TSDCE) remain advantageous at high SNR, both ResNet and U-Net are competitive, with U-Net frequently matching the performance of DFT-based channel estimation algorithm (DFT-CEA). The U-Net also demonstrated a tighter distribution of estimation errors in medium-to-high SNR regimes, which is desirable for reliability. Additionally, the proposed schemes showed robustness to variations in the number of multipath components and were resilient to realistic phase-shifting hardware impairments.

We also conducted a detailed complexity analysis, which confirmed that the proposed schemes remain computationally viable for large-scale MIMO settings, with asymptotic complexities comparable to those of traditional methods under practical assumptions. Future work will explore further optimization and training using more realistic datasets collected from real-world 6G network deployments. Another promising direction is the use of complex-valued neural networks, which may better exploit the structure of complex-valued observations and further improve estimation accuracy. We also plan to extend our method to planar array configurations for 2D angle estimation (azimuth and elevation), potentially leveraging tensor-based or multi-dimensional learning architectures.

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